

Generation Interconnection System Impact Study Report

For Queue Project Q17-0825

Cayucos 138 kV
September 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0825. The Interconnection Customer proposes a 128.3 MW solar photovoltaic facility interconnecting at the Cayucos 138 kV substation in the Pennsylvania area. The request was received on July 12, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 357 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,998,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Big Hollow Road Substation / Mesa Rim 138 KV	W Camus / Edes Tap 2 345 KV	165.2	158	7.7
Key est / Big Hollow Road 138 KV	Edes Tap 2 / West Campus 345 KV	156.0	198	5.5
Ke West / Mid America 138 KV Line	West Campus / Edes Tap 2 345 KV	145.7	214	5.5
Mesa Rim / Phillippi 138 KV	West Campus / Edes Tap 2 345 KV line	122.6	221	7.7
Huntingdon / Phillippi 138 KV	W Campus / Edes Tap 2 345 KV	114.6	244	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Hollow Road Substation / Mesa Rim 138 KV	Rebuild line with higher-capacity conductor	\$1,870,000	100% (\$1,870,000)
Key est / Big Hollow Road 138 KV	Rebuild line with higher-capacity conductor	\$2,553,000	100% (\$2,553,000)
Key W / Mid America 138 KV	Rebuild line with higher-capacity conductor	\$4,714,000	100% (\$4,714,000)
Phillippi / Mesa Rim 138 KV Line	Rebuild line with higher-capacity conductor	\$6,463,000	100% (\$6,463,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Huntingdon / Phillippi 138 KV	Rebuild line with higher-capacity conductor	\$6,398,000	100% (\$6,398,000)
	Total allocated to project		\$21,998,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.